MATH 130: 10/22 WORKSHEET

We've been looking at numbers which summarize the spread of a data set, as the five number summary of minimum, first quartile, median, third quartile, and maximum. It would be nice if there were a single number to summarize the spread of a data set. While a single number could never tell us everything about a data set, there is a single measure of spread that is mathematically useful.

Here's the intuitive idea first. First identify the center of your data set. Next determine the average distance of the data to this center. This average gives a measure of how far from the center the data are. A small average means they are tightly clustered, while a large average means they are more spread out from the center.

The precise definition takes this idea, but for technical reasons it's better to look at the average of the *squares* of the distances.

Standard deviation.

Consider a data set

$$x_1, x_2, \ldots, x_n$$

with mean $\bar{x}$. The *variance* Var is the average of the squares of the distances from the mean, and the *standard deviation* s is the square root of the variance. As formulas:

$$\text{Var} = \frac{\sum (x_i - \bar{x})^2}{n} = \frac{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \cdots + (x_n - \bar{x})^2}{n}$$

$$s = \sqrt{\text{Var}} = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n}}$$

The standard deviation is the more important of the two concepts. One way you can think about it is, the variance is an intermediate step in calculating the standard deviation; first calculate the variance, then take its square root.

Warning! Be careful with using software to check your work. Many programs will use a slightly different formula for their standard deviation function. Namely, they will divide by $n - 1$ instead of dividing by n. The reasons for the different formula are technical ("Bessel's correction" is the keyword to search); briefly, if you are trying to estimate an unknown standard deviation based on a sample you want to use the corrected version of the formula.